

4.5.4 Binary number system

4.5.4.1 Unsigned binary

Content	Additional information
Know the difference between unsigned binary and signed binary.	Students are expected to be able to convert between unsigned binary and decimal and vice versa.
Know that in unsigned binary the minimum and maximum values for a given number of bits, n , are 0 and $2^n - 1$ respectively.	

4.5.4.2 Unsigned binary arithmetic

Content	Additional information
Be able to: • add two unsigned binary integers • multiply two unsigned binary integers.	

4.5.4.3 Signed binary using two's complement

Content	Additional information
Know that signed binary can be used to represent negative integers and that one possible coding scheme is two's complement.	This is the only representation of negative integers that will be examined. Students are expected to be able to convert between signed binary and decimal and vice versa.
Know how to: • represent negative and positive integers in two's complement • perform subtraction using two's complement • calculate the range of a given number of bits, n.	

4.5.4.4 Numbers with a fractional part

Content	Additional information
Know how numbers with a fractional part can be represented in: • fixed point form in binary in a given number of bits • floating point form in binary in a given number of bits.	Students are not required to know the Institute of Electrical and Electronic Engineers (IEEE) standard, only to know, understand and be able to use a simplified floating representation consisting of mantissa + exponent.
Be able to convert for each representation from: • decimal to binary of a given number of bits • binary to decimal of a given number of bits.	Exam questions on floating point numbers will use a format in which both the mantissa and exponent are represented using two's complement.

4.5.4.5 Rounding errors

Content	Additional information
Know and be able to explain why both fixed point and floating point representation of decimal numbers may be inaccurate.	Use binary fractions. For a real number to be represented exactly by the binary number system, it must be capable of being represented by a binary fraction in the given number of bits. Some values cannot ever be represented exactly, for example 0.1_{10} .

4.5.4.6 Absolute and relative errors

Content	Additional information
Be able to calculate the absolute error of numerical data stored and processed in computer systems.	
Be able to calculate the relative error of numerical data stored and processed in computer systems.	
Compare absolute and relative errors for large and small magnitude numbers, and numbers close to one.	

4.5.4.7 Range and precision

Content	Additional information
Compare the advantages and disadvantages of fixed point and floating point forms in terms of range, precision and speed of calculation.	

4.5.4.8 Normalisation of floating point form

Content	Additional information
Know why floating point numbers are normalised and be able to normalise un-normalised floating point numbers with positive or negative mantissas.	

4.5.4.9 Underflow and overflow

Content	Additional information
Explain underflow and overflow and describe the circumstances in which they occur.	